

if you want to buy a plot of land on Decentraland, you will have to convert your real-world currency to MANA, the native token of the game. It's the same story for all the games and platforms on the metaverse.

Augmented reality and virtual

reality: Augmented reality (AR) and virtual reality (VR) engines play an important role in enabling the metaverse to be an enthralling virtual experience. They help create immersive and engaging 3D environments for the metaverse.

While VR and the metaverse might seem like the same thing, there are certain differences between the two:

- VR is only a part of the broader metaverse. The metaverse is made up of several other technologies that complement VR technology.
- While VR enables users to see 3D simulations, the problem is that VR technologies cannot provide physical stimulations, which is an important part of the metaverse.

This is where AR comes in. It helps expand the scope of the metaverse by enabling users to actually feel things by creating physical stimulations. As a result, users can hear, feel and interact with the virtual metaverse as if they are physically present in it. The combination of VR and AR technologies is likely to create a more realistic metaverse and bring in massive investments from various companies.

Artificial intelligence (AI): AI has become an inextricable part of our lives — from Alexa and Siri to the smart sensing air conditioner that helps us sleep. So, it is only expected that the metaverse will also use AI to improve its performance.

AI plays a major role in making every process that happens in the metaverse an immersive one. It can help process data at lightning speed with the aid of machine learning techniques. Within the metaverse,

prior historical data can be used by AI algorithms to create unique outputs and insights. Another important use of AI in the metaverse is to create metaverse avatars. AI engines can analyse 2D and 3D images to generate more realistic looking and accurate avatars. AI can also add more character by helping create different facial expressions, hair styles, clothes, and features to enhance the appearance of the avatars.

Internet of Things (IoT): IoT enables applications to collect data and feed it to the metaverse to adjust more precisely to real-world conditions. For example, weather or temperature data collected via IoT can help the metaverse adjust its environment.

Another advantage of IoT is that it can seamlessly connect the 3D world to a large number of devices from the physical world, enabling real-time stimulations in the metaverse.

IoT can also use AI and machine learning algorithms to further improve on and optimise the metaverse environment by managing the data that it collects.

Edge computing and 5G

communications: Edge computing basically facilitates faster data transfer with fewer delays and helps enable smoother experiences. It is one of the primary factors that has enabled the metaverse to be viable, as computers need to handle the intense stimulations with efficiency to reduce lag and ensure that users have an immersive and unhindered gaming experience.

Another important feature that is often combined with edge computing is the availability of 5G networks. Previously, users often faced issues with their network while engaging in the metaverse due to slow data transfer speed resulting from unavailability of fast internet speed. However, with 5G being rolled out and made available at affordable rates, users can experience the metaverse from their desktops and other devices without facing a lag due to network speed.

Together, edge computing and 5G enable users to engage in the metaverse and enjoy a truly immersive experience. **END** 🐼

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